

Linear Algebra Recitation Class: Week 1

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September 10, 2026

Throughout these notes, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ denote the sets of rational, real, and complex numbers, respectively.

Definition 1. Let F be $\mathbb{Q}$ or $\mathbb{R}$ or $\mathbb{C}$. A **vector space** over F (or an F vector space) is a nonempty set V together with two operations:

$$\begin{aligned} (\text{vector addition}) : V \times V &\longrightarrow V & (\text{scalar multiplication}) : F \times V &\longrightarrow V \\ (v_1, v_2) &\longmapsto v_1 + v_2 & (a, v) &\longmapsto a \cdot v \end{aligned}$$

satisfying:

VS1. For any $v_1, v_2 \in V$, $v_1 + v_2 = v_2 + v_1$.

VS2. For any $v_1, v_2, v_3 \in V$, $v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$.

VS3. There exists an element $0_V \in V$ such that $v + 0_V = v = 0_V + v$ for any $v \in V$.

VS4. For any $v \in V$, there exists an element $v' \in V$ such that $v + v' = 0_V$.

VS5. For any $v \in V$, $1 \cdot v = v$. (Note that $1 \in F$.)

VS6. For any $a, b \in F$, $v \in V$, $(a \cdot_F b) \cdot v = a \cdot (b \cdot v)$.

VS7. For any $a \in F$, $v_1, v_2 \in V$, $a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$.

VS8. For any $a, b \in F$, $v \in V$, $(a +_F b) \cdot v = (a \cdot v) + (b \cdot v)$.

Here $+_F$ and $\cdot_F$ denote addition and multiplication in F , respectively.

Example. The vector addition and the scalar multiplication of a vector space need NOT be those we normally use. For example, consider the set $\mathbb{R}^+$ of all positive real numbers, i.e.

$$\mathbb{R}^+ := \{r \in \mathbb{R} \mid r > 0\}.$$

For any $v_1, v_2, v \in \mathbb{R}^+$, $q \in \mathbb{Q}$, we define

$$v_1 \boxplus v_2 := v_1 v_2 \text{ (standard multiplication)} \quad q \boxtimes v := v^q \text{ (standard exponentiation)}.$$

Then $\mathbb{R}^+$ is a vector space over $\mathbb{Q}$. Let's check this directly:

VS1. Given $v_1, v_2 \in \mathbb{R}^+$, $v_1 \boxplus v_2 = v_1 v_2 = v_2 v_1 = v_2 \boxplus v_1$. ✓

VS2. Given $v_1, v_2, v_3 \in \mathbb{R}^+$,

$$v_1 \boxplus (v_2 \boxplus v_3) = v_1 \boxplus (v_2 v_3) = v_1 v_2 v_3 = (v_1 v_2) \boxplus v_3 = (v_1 \boxplus v_2) \boxplus v_3. \checkmark$$

VS3. Define $0_{\mathbb{R}^+} := 1 \in \mathbb{R}^+$, then given $v \in \mathbb{R}^+$, we have

$$v \boxplus 0_{\mathbb{R}^+} = v \cdot 1 = v = 1 \cdot v = 0_{\mathbb{R}^+} \boxplus v. \checkmark$$

VS4. Given $v \in \mathbb{R}^+$, we define $v' := 1/v \in \mathbb{R}^+$, then

$$v \boxplus v' = v \cdot 1/v = 1 = 0_{\mathbb{R}^+}. \checkmark$$

VS5. Given $v \in \mathbb{R}^+$, $1 \boxtimes v = v^1 = v$. $\checkmark$ (Note that the 1 here is the 1 in $\mathbb{Q}$)

VS6. Given $a, b \in \mathbb{Q}$, $v \in \mathbb{R}^+$,

$$(a \cdot_{\mathbb{Q}} b) \boxtimes v = v^{ab} = (v^b)^a = a \boxtimes v^b = a \boxtimes (b \boxtimes v). \checkmark$$

VS7. Given $a \in \mathbb{Q}$, $v_1, v_2 \in \mathbb{R}^+$,

$$a \boxtimes (v_1 \boxplus v_2) = a \boxtimes (v_1 v_2) = (v_1 v_2)^a = v_1^a v_2^a = v_1^a \boxplus v_2^a = (a \boxtimes v_1) \boxplus (a \boxtimes v_2). \checkmark$$

VS8. Given $a, b \in \mathbb{Q}$, $v \in \mathbb{R}^+$,

$$(a +_{\mathbb{Q}} b) \boxtimes v = v^{a+b} = v^a v^b = v^a \boxplus v^b = (a \boxtimes v) \boxplus (b \boxtimes v). \checkmark$$

In the rest of these notes, we introduce some algebraic structures that are studied in an abstract algebra course.

Definition 2. A **group** is a nonempty set G with a binary operation

$$\begin{aligned} \cdot : G \times G &\longrightarrow G \\ (g_1, g_2) &\longmapsto g_1 \cdot g_2 \end{aligned}$$

satisfying:

G1. For any $g_1, g_2, g_3 \in G$, $g_1 \cdot (g_2 \cdot g_3) = (g_1 \cdot g_2) \cdot g_3$.

G2. There exists an element $e \in G$ such that $g \cdot e = g = e \cdot g$ for any $g \in G$.

G3. For any $g \in G$, there exists $g' \in G$ such that $g \cdot g' = e = g' \cdot g$.

If, in addition, G satisfies

Ga. For all $g_1, g_2 \in G$, $g_1 \cdot g_2 = g_2 \cdot g_1$,

then we say G is **abelian** (or G is an abelian group), named after a Norwegian mathematician NIELS HENRIK ABEL.

Remark. If G is a group, it can be proved that the element e defined in G2 is unique, which is called the **identity** of G . Given element $g \in G$, the element g' in G3 is also unique, which is called the **inverse** of g .

Example. The set of integers $\mathbb{Z}$ with the standard addition $+$ is an abelian group. The identity element of $\mathbb{Z}$ is 0 and Given $z \in \mathbb{Z}$, the inverse of z is $-z$.

Example (General linear groups). The set of all 2×2 invertible matrices with coefficients in $\mathbb{R}$

$$\text{GL}_2(\mathbb{R}) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : ad - bc \neq 0, \quad a, b, c, d \in \mathbb{R} \right\}$$

with matrix multiplication forms a group. It is NOT abelian. For example,

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Example. If V is a vector space over F , the axioms [VS1] to [VS4] say precisely that V is an abelian group under vector addition.

Exercise A. Give an example of an abelian group and an example of a non-abelian group, other than those mentioned above.

Definition 3. A **ring** is a nonempty set R with two binary operations

$$\begin{aligned} + : R \times R &\longrightarrow R & \cdot : R \times R &\longrightarrow R \\ (r_1, r_2) &\longmapsto r_1 + r_2 & (r_1, r_2) &\longmapsto r_1 \cdot r_2 \end{aligned}$$

satisfying

Ra. R is an abelian group with $+$, the identity element is denoted by 0_R

R1. For any $r_1, r_2, r_3 \in R$, $r_1 \cdot (r_2 \cdot r_3) = (r_1 \cdot r_2) \cdot r_3$.

R2. There exists an element $1_R \in R$, such that $r \cdot 1_R = r = 1_R \cdot r$ for any $r \in R$.

R3. For any $r_1, r_2, r_3 \in R$, $r_1 \cdot (r_2 + r_3) = r_1 \cdot r_2 + r_1 \cdot r_3$.

R4. For any $r_1, r_2, r_3 \in R$, $(r_1 + r_2) \cdot r_3 = (r_1 \cdot r_3) + (r_2 \cdot r_3)$.

If R furthermore satisfies

Rc. For any $r_1, r_2 \in R$, $r_1 \cdot r_2 = r_2 \cdot r_1$.

Then we say R is **commutative**, or R is a commutative ring.

Remark. As in the case of groups, the elements 0_R and 1_R are unique. They are called the **additive identity** and **multiplicative identity** of R , respectively. Regarding R as an abelian group under addition, each $r \in R$ has a unique additive inverse, denoted by $-r$.

Example. $\mathbb{Z}$ is a commutative ring equipped with standard addition and multiplication, where $0_{\mathbb{Z}} = 0$, $1_{\mathbb{Z}} = 1$.

Example (Matrix rings). The set of all 2×2 matrices with coefficients in $\mathbb{R}$

$$R := M_2(\mathbb{R}) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R} \right\}$$

with matrix addition and matrix multiplication forms a ring, where 0_R is the zero matrix, and 1_R is the identity matrix. It is NOT commutative (see the counterexample of $\text{GL}_2(\mathbb{R})$ above).

Remark. The ring $M_2(\mathbb{R})$ is also an **$\mathbb{R}$ -algebra**. Roughly speaking, an $\mathbb{R}$ -algebra is both a ring and a vector space over $\mathbb{R}$, with suitable compatibility conditions between multiplication and scalar multiplication.

Example (Polynomial rings). The set of polynomials with coefficients in $\mathbb{R}$

$$\mathbb{R}[x] := \{f(x) \mid f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \quad a_0, a_1, \dots, a_n \in \mathbb{R}, n \geq 0\}$$

forms a commutative ring with standard polynomial addition and polynomial multiplication. The additive identity of $\mathbb{R}[x]$ is the zero polynomial 0 and the multiplicative identity is the constant polynomial 1.

Remark. Similarly, $\mathbb{R}[x]$ is a commutative $\mathbb{R}$ -algebra, i.e. an $\mathbb{R}$ -algebra which is also a commutative ring.

Exercise B. Give an example of a commutative ring and an example of a non-commutative ring, other than those mentioned above.

Both $\mathbb{Z}$ and $\mathbb{Q}$ are commutative rings under the usual addition and multiplication. However, every nonzero rational number $q = a/b$ has a multiplicative inverse $q^{-1} = b/a \in \mathbb{Q}$. In contrast, most nonzero integers do not have multiplicative inverses in $\mathbb{Z}$. This is the very difference between a general commutative ring and a field.

Definition 4. A **field** F is a commutative ring with $1_F \neq 0_F$ where every nonzero element has a multiplicative inverse.

Remark. If we do not require F to be commutative, then F is called a **division ring** or a **skew field**.

Exercise C. Check that $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are all fields. Prove that

$$\mathbb{Q}(\sqrt{2}) := \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$$

is a field.

Exercise D. If F, K are fields and $F \subseteq K$, we say F is a **subfield** of K . Prove that if F is a subfield of K , then K is a vector space over F . This is a basic yet really important fact in field theory.

Lastly, we introduce an algebraic structure similar to a vector space over a field.

Definition 5. Let R be a ring, a **left R -module** or a module over R is a nonempty set M together with two operations:

$$\begin{aligned} + : M \times M &\longrightarrow M & \cdot : R \times M &\longrightarrow M \\ (m_1, m_2) &\longmapsto m_1 + m_2 & (a, m) &\longmapsto a \cdot m \end{aligned}$$

satisfying:

M1. For any $m_1, m_2 \in M$, $m_1 + m_2 = m_2 + m_1$.

- M2. For any $m_1, m_2, m_3 \in M$, $m_1 + (m_2 + m_3) = (m_1 + m_2) + m_3$.
- M3. There exists an element $0_M \in M$ such that $m + 0_M = m = 0_M + m$ for any $m \in M$.
- M4. For any $m \in M$, there exists an element $m' \in M$ such that $m + m' = 0_M$.
- M5. For any $m \in M$, $1_R \cdot m = m$.
- M6. For any $a, b \in R$, $m \in M$, $(a \cdot_R b) \cdot m = a \cdot (b \cdot m)$.
- M7. For any $a \in R$, $m_1, m_2 \in M$, $a \cdot (m_1 + m_2) = a \cdot m_1 + a \cdot m_2$.
- M8. For any $a, b \in R$, $m \in M$, $(a +_R b) \cdot m = (a \cdot m) + (b \cdot m)$.

Here $+_R$ and $\cdot_R$ denote addition and multiplication in R , respectively.

Remark. One can similarly define a right R -module. If R is commutative, every left R -module can be seen as a right R -module by defining $m \cdot r := r \cdot m$. Note that the module axioms are analogous to the vector-space axioms. Therefore, one may think of an R -module as a "vector space over a ring."

Exercise E. If F is a field (which is also a ring by definition), convince yourself that a left F -module is a right F -module and a vector space over F .

Example. Any abelian group G with binary operation $+$ with identity 0_G can be seen as a $\mathbb{Z}$ -module via.

$$+ : G \times G \longrightarrow G \qquad \cdot : \mathbb{Z} \times G \longrightarrow G$$

$$(g_1, g_2) \longmapsto g_1 + g_2 \qquad (n, g) \longmapsto n \cdot g = \begin{cases} \underbrace{g + g + \cdots + g}_{n \text{ times}} & \text{if } n > 0 \\ 0_G & \text{if } n = 0 \\ \text{inverse of } |n| \cdot g & \text{if } n < 0 \end{cases}$$

Example. Let $M = \mathbb{R}^2 = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} : a, b \in \mathbb{R} \right\}$, $R = M_2(\mathbb{R})$, then M is a left R -module with addition and multiplication defined by

$$+ : M \times M \longrightarrow M \qquad \cdot : R \times M \longrightarrow M$$

$$\left(\begin{pmatrix} a_1 \\ b_1 \end{pmatrix}, \begin{pmatrix} a_2 \\ b_2 \end{pmatrix} \right) \longmapsto \begin{pmatrix} a_1 + a_2 \\ b_1 + b_2 \end{pmatrix} \qquad \left(\begin{pmatrix} s & t \\ u & v \end{pmatrix}, \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} \right) \longmapsto \begin{pmatrix} s & t \\ u & v \end{pmatrix} \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$$

Exercise F. Check (or convince yourself) that these two examples have module structures.

References

- [FIS03] Stephen H. Friedberg, Arnold J. Insel, and Lawrence E. Spence. *Linear algebra*. Prentice Hall, Inc., Upper Saddle River, NJ, fourth edition, 2003.
- [Fra02] John B. Fraleigh. *A first course in abstract algebra*. Addison-Wesley Publishing Co., Reading, Mass.-London-Don Mills, Ont., 2002.